

Boao (James) Chen

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RESEARCH INTERESTS

Databases, Data Systems, Storage Engines, Machine Learning for Systems

EDUCATION

Doctor of Philosophy, Computer Science

Brandeis University, Waltham, MA

Advisor: Subhadeep Sarkar

Sep 2024 – Present

Bachelor of Arts, Computer Science

Cornell University, Ithaca, NY

Washington University, St. Louis, MO

Aug 2020 – May 2024

PUBLICATIONS

In Review	Boao Chen , Shubham Kaushik, and Subhadeep Sarkar. <i>Mind the Buffer: Dissecting the LSM-Buffer Design Space</i> , in Review, 2026.
TPCTC 2025	Alexander H. Ott, Shubham Kaushik, Boao Chen , and Subhadeep Sarkar. <i>Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking</i> , Performance Evaluation and Benchmarking TPC Technology Conference.
MobiCom 2024	Ke Li, Ruidong Zhang, Boao Chen , Siyuan Chen, Sicheng Yin, Saif Mahmud, Francois Guimbretiere, and Cheng Zhang. <i>GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame</i> , International Conference on Mobile Computing and Networking.
CHI 2024	Ke Li, Ruidong Zhang, Siyuan Chen, Boao Chen , Mose Sakashita, Francois Guimbretiere, and Cheng Zhang. <i>EyeEcho: Continuous Facial Expression Tracking on Glasses with Minimally-obtrusive Acoustic Sensing</i> , ACM Computer Human Interaction Conference.
IMWUT 2024	Ke Li, Devansh Agarwal, Ruidong Zhang, Vipin Gunda, Tianjun Mo, Saif Mahmud, Boao Chen , Francois Guimbretiere, and Cheng Zhang. <i>SonicID: User Identification on Smart Glasses with Acoustic Sensing</i> , The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.

POSTER

NEDB 2026	Shubham Kaushik, Steven Yang, Boao Chen , and Subhadeep Sarkar. <i>FluidLSM: A Fluid Log-Structured Merge Tree for Shifting Workloads</i> , North East Database Day.
NEDB 2026	Alexander Ott, Shubham Kaushik, Boao Chen , and Subhadeep Sarkar. <i>Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking</i> , North East Database Day.
NEDB 2025	Alexander Ott, Boao Chen , Arpita Saha, and Subhadeep Sarkar. <i>Self-Designing LSM Memory Buffer</i> , North East Database Day.

TECHNICAL SKILLS

Programming Languages: C, C++ , Java, Python, SQL, JavaScript, Node.js, R, MATLAB

Databases: RocksDB, MariaDB, Postgres, Cassandra, MongoDB

Machine Learning: PyTorch, TensorFlow, keras

RESEARCH PROJECTS

- Mind the Buffer** (*Under Review*, 2026): Conducted a comprehensive analysis of the LSM-tree in-memory buffer design space—the critical component through which every read and write operation must pass—evaluating how buffer implementation, sizing, and tuning bottleneck NoSQL storage engine performance across shifting workloads and hardware. Benchmarked 11 distinct buffer designs—including three hash-hybrid variants—and distilled the empirical tradeoffs into a practitioner handbook of 10 actionable guidelines for buffer selection and tuning. Designed and evaluated an adaptive buffer that dynamically switches implementations on-the-fly to sustain peak performance across workload shifts.
- Tectonic+** (*Ongoing*): Extending Tectonic into a next-generation benchmarking suite intended to replace YCSB and other static workload generators for key-value stores. Tectonic+ builds on Tectonic’s workload-modeling core while

adding support for concurrency and multiple storage backends (RocksDB, Cassandra, Redis, ScyllaDB) through a unified database abstraction layer.

- **Tectonic** (*Published, TPCTC 2025*): Co-developed a configurable, resource-efficient Rust-based key-value workload generator to model the temporal, structural, and dynamic properties of production workloads missing in benchmarks like YCSB and KVBench. Implemented fine-grained control over multi-operation access patterns (point/range queries, merges, and deletes), prefix-distributed composite key generation, and user-specified data sortedness, achieving 2× higher throughput and up to an 84% lower main memory footprint compared to state-of-the-art generators.

TEACHING ASSISTANT

Spring 2026	Introduction to Problem Solving in Python (COSI 10a), Brandeis University
Fall 2025	Capstone Project for Software Engineering (COSI 166B), Brandeis University
Spring 2025	Database Management Systems (COSI 127B), Brandeis University
Fall 2024	Algorithms (COSI 180A), Brandeis University
Spring 2024	Applied High-Performance and Parallel Computing (CS 5520), Cornell University
Fall 2023	Foundations of Artificial Intelligence (CS 4700), Cornell University
Spring 2023	Operating Systems (CS 4410), Cornell University

PROFESSIONAL SERVICES

2026	Student Volunteer for VLDB 2026
2026	Web Chair for Northeast Database (NEDB) Day 2026
2026 – Present	Member, Institute of Electrical and Electronics Engineers (IEEE)
2025 – Present	Member, Association for Computing Machinery (ACM)

CERTIFICATION

Jun 2025	Oracle Certified Professional, MySQL 8.0 Database Administrator – [link]
Jun 2025	Oracle Certified Associate, MySQL Implementation – [link]